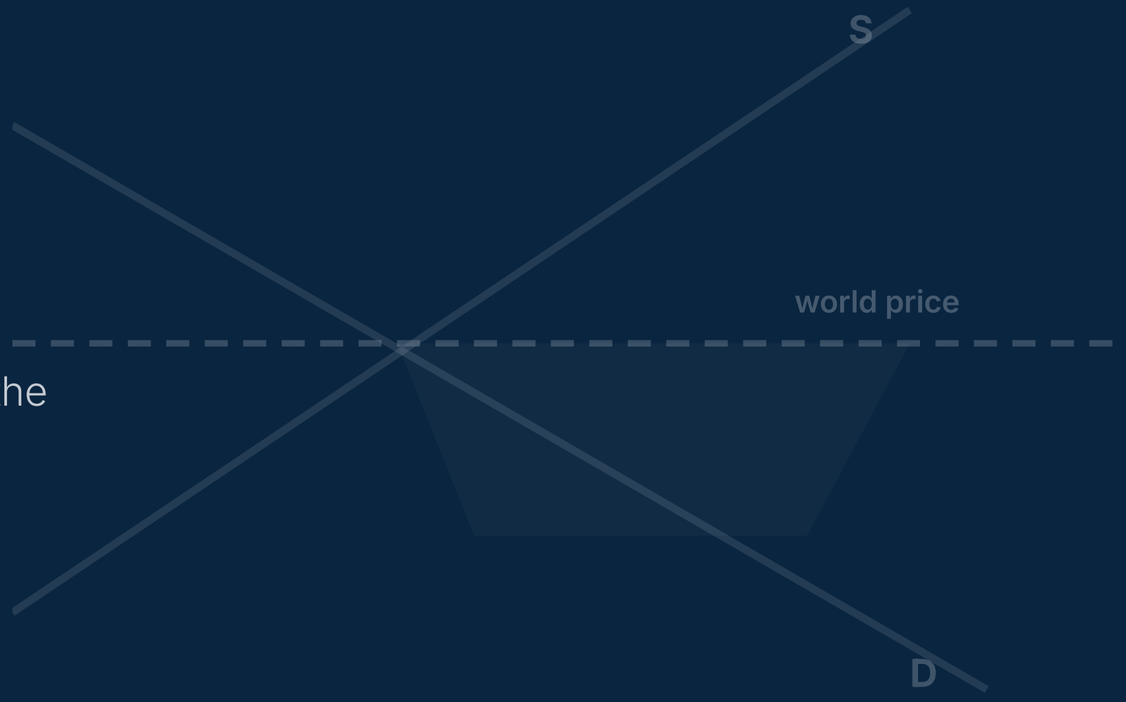
AF3625 · ENGINEERING ECONOMICS · TUTORIAL PRESENTATION

Trade barriers and welfare

Import quotas, tariffs and deadweight loss in a small open economy — the Group 5 set, Questions 5.1 to 5.4.



GROUP	PRESENTERS	SLOT	RUNTIME
Group 5	add member names	Week 9 · 27–30 Oct 2026	≈ 22 minutes

ROADMAP

What we will cover

01 · Primer

Small country, world price, consumer and producer surplus, and the two ways of restricting trade.

02 · Q5.1 TV market

Why country Z imports, how much, producer surplus, and the effect of a 20,000-unit import quota.

03 · Q5.2 Handbags

Algebra: autarky, free trade, the tariff that cuts imports by 25%, revenue and consumer surplus.

04 · Q5.3 Egypt

Generators: the post-tariff price, domestic output, imports, tariff revenue, gainers and losers.

05 · Q5.4 Article

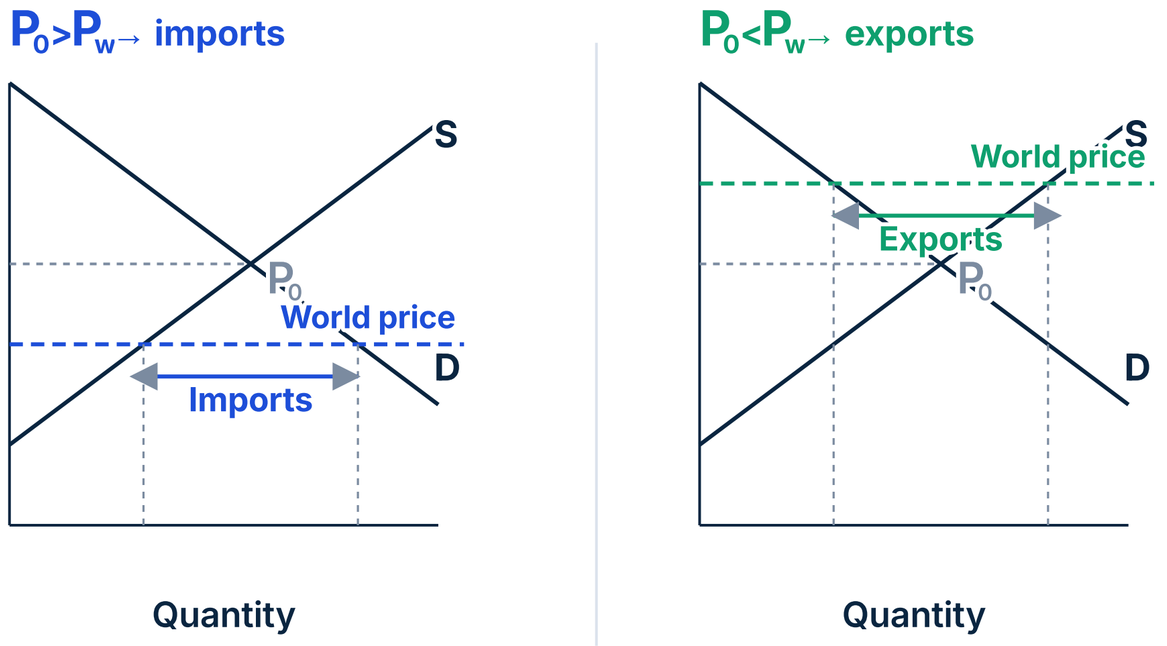
The EU’s 2024 duties on Chinese electric cars, read through Topic 5.

06 · Conclusion

What the four questions have in common — and the Q&A.

A small country takes the world price as given

FIG. 1 · THE IMPORT / EXPORT RULE



Left: the autarky price is above the world price, so at that world price domestic demand exceeds domestic supply — the gap is imported. Right: the opposite case, where the country exports.

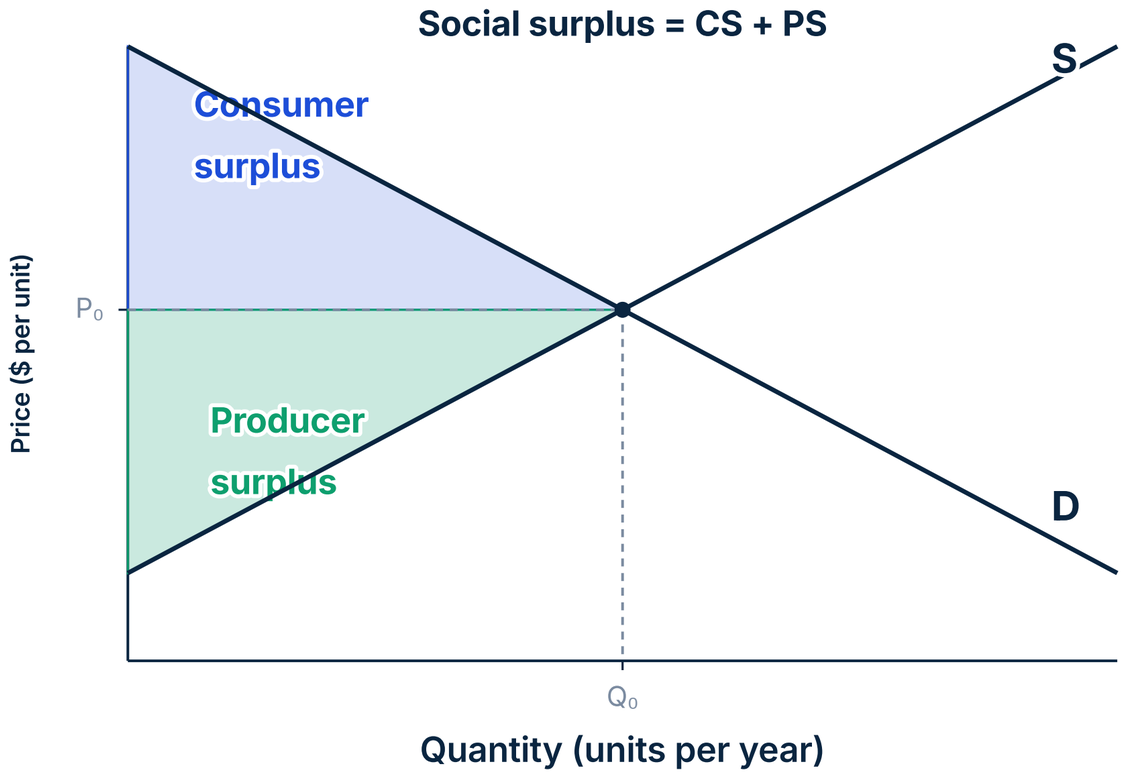
- A **small country** is a **price taker**: its own demand and supply cannot move the world price, so the world price is a **horizontal line** at P_w.
- Once the country trades, domestic buyers and sellers all face P_w.
- Autarky price $P_0 > P_w$ → the country **imports**; $P_0 < P_w$ → it **exports**.

$$Q_{import} = Q_d(P_w) - Q_s(P_w)$$

The gap between what domestic consumers want and what domestic producers supply at the world price is filled by imports.

Surplus, and the two ways to restrict trade

FIG. 2 · CONSUMER, PRODUCER AND SOCIAL SURPLUS



Social surplus — the whole shaded area — is largest at the autarky equilibrium.

consumer surplus producer surplus

	TARIFF	IMPORT QUOTA
Instrument	tax on each imported unit	ceiling on the import *quantity*
Domestic price	$P_W + t$	rises until imports = the quota
Supply curve	unchanged domestic *S*	S + quota
Transfer goes to	**the government (tariff revenue)**	**importers (quota rent)**
Cost to society	**two deadweight-loss triangles**	**the same two triangles**

The television market in country Z

5.1

QUESTION 5.1

Suppose country Z is a small country where all buyers and sellers there are price takers. The figure below shows the domestic demand and domestic supply curves of television (TV) market in country Z. The country is open for international trade.

i) Under international free trade, explain why country Z will import televisions.

ii) Calculate the number of televisions that country Z will import under international free trade.

iii) What is the amount of producer surplus in country Z under international free trade? Show all the steps of calculation.

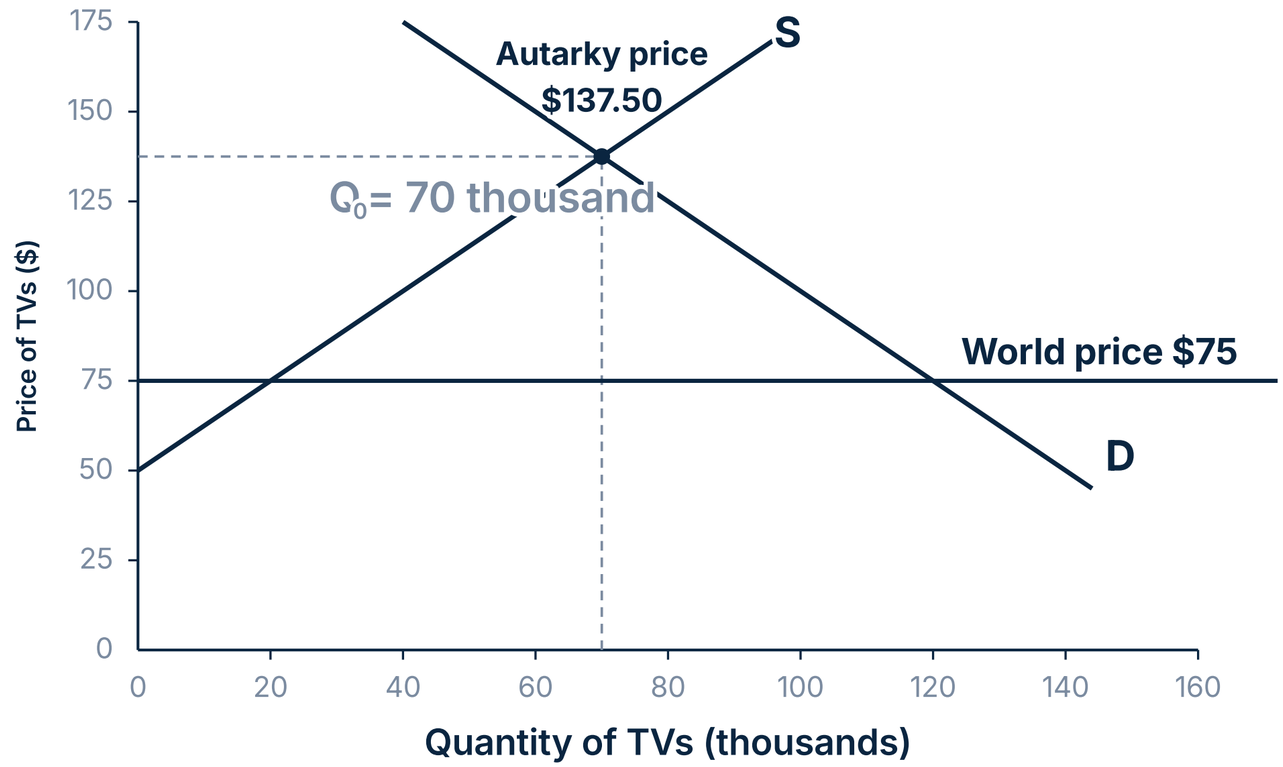
iv) Country Z government intervenes in international trade by imposing an import quota of 20,000 televisions. State the intended effect of such government intervention.

v) The “Domestic supply plus quota” curve is not plotted correctly in the diagram. Plot the correct domestic supply curve plus quota (label it as S + quota) in the diagram above.

vi) What is the amount of quota rent? Show all the steps of calculation.

Reading the diagram

FIG. 3 · COUNTRY Z: AUTARKY AND THE WORLD PRICE



Read off the figure: D is P = 225 − 1.25Q and S is P = 50 + 1.25Q (Q in thousands).

Read the two straight lines off the figure (Q in thousands of TVs):

```
Demand:  P = 225 - 1.25 Q
```

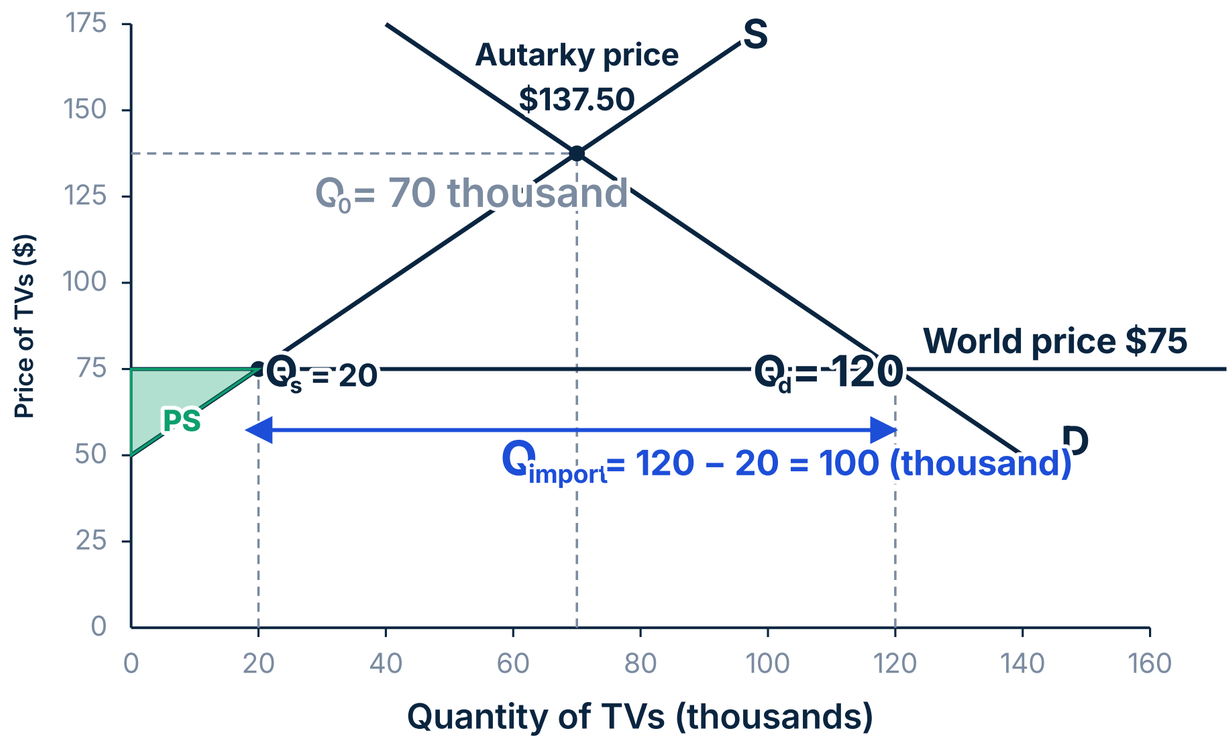
```
Supply:  P = 50 + 1.25 Q
```

- **Autarky** ($D = S$): $Q_0 = 70{,}000$ and $P_0 = \$137.50$.
- **World price = $75** — a horizontal line, because Z is a small country and cannot influence it.
- The world price is **below** the autarky price, so at $75 domestic consumers want more than domestic producers supply → **imports**.

QUESTION 5.1 (I) AND (II)

Why Z imports — and how many TVs

FIG. 4 · FREE TRADE: IMPORTS = Q_D - Q_S



At $75 domestic producers supply 20,000 TVs and consumers demand 120,000 — the gap is imported.

(i) Why imports?

- At $75 producers supply **20,000** but consumers want **120,000** — and foreign producers will sell at $75, so the shortfall is imported.

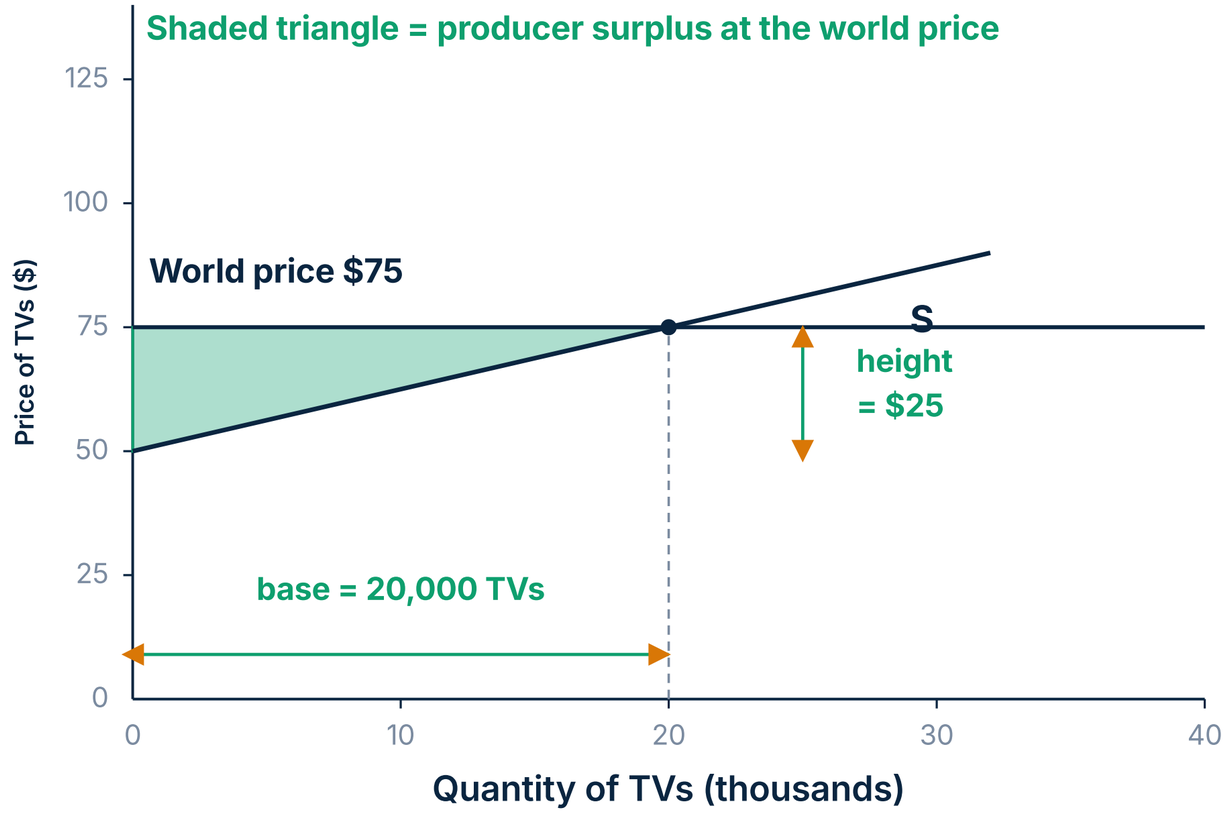
(ii) How many?

1. At $P = \$75$: $Q_s = 20$ thousand, $Q_d = 120$ thousand.
2. Imports = $Q_d - Q_s = 100$ thousand = **100,000 TVs**.

QUESTION 5.1 (III)

Producer surplus under free trade

FIG. 5 · PRODUCER SURPLUS AT THE WORLD PRICE



½ × 20,000 × $25 = $250,000. Only domestic production counts, so the base is 20,000, not 120,000.

Producer surplus is the area **above** the domestic supply curve and **below** the price, up to the quantity sold. At the world price it is a triangle.

1. Base = quantity supplied at $75 = **20,000** TVs.
2. Height = price – the price at which supply starts = `$75 - $50 = $25`.
3. PS = ½ × base × height = ½ × 20,000 × $25

> PS = **$250,000**

Units: 20,000 TVs × $25 ÷ 2 → dollars.

QUESTION 5.1 (IV)

What a 20,000-unit import quota is meant to do

1 · Restrict supply

Legal imports are capped at 20,000 TVs — far below the 100,000 that would be imported under free trade, so the quota **binds**.

2 · Raise the domestic price

With less total supply available, consumers bid the domestic price **above** the world price; domestic producers respond by increasing output.

3 · Protect domestic producers

Higher prices and larger domestic sales raise producer surplus, but consumers pay more and their surplus falls.

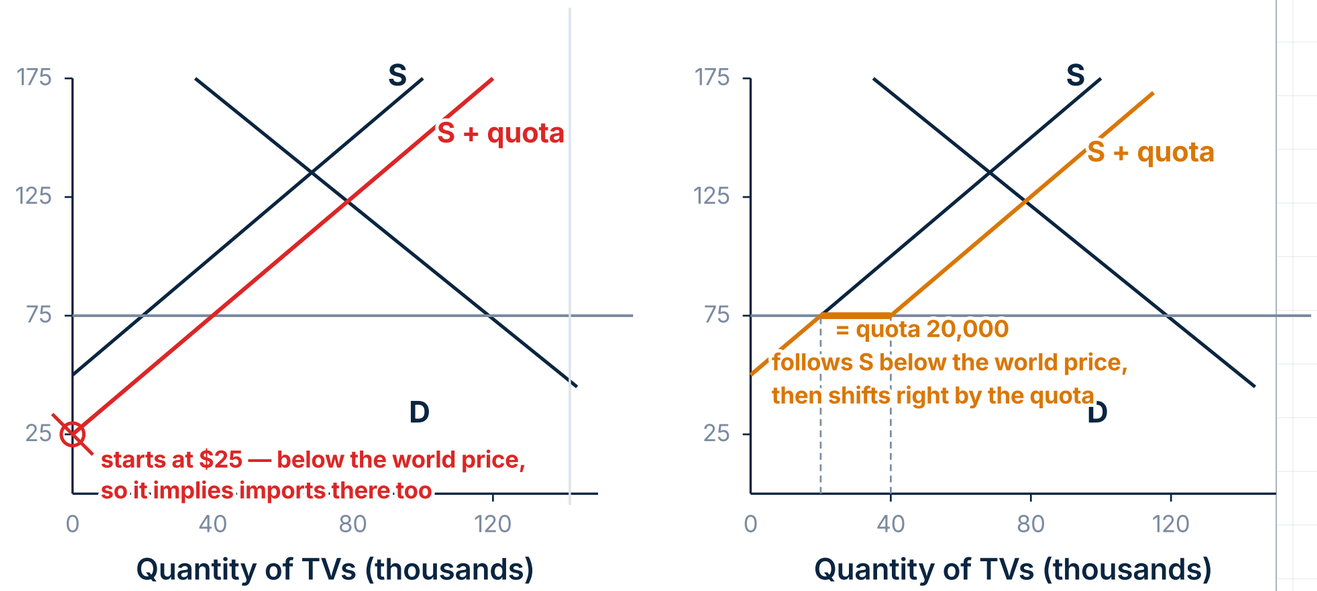
Intended effect, in one sentence:

By limiting imports to 20,000 TVs, the government intends to **reduce the quantity of imports, raise the domestic price and domestic production, and thereby protect domestic TV producers from foreign competition**.

20k	100k	-80%
QUOTA (IMPORTS)	FREE-TRADE IMPORTS	REDUCTION IN IMPORTS

Plotting “S + quota” correctly

FIG. 7 · AS PRINTED (RED) AND AS IT SHOULD BE (AMBER)



Left: the printed curve, one straight line from $25. Right: kinked — domestic S below the world price, then shifted right by the quota.

The rule from Topic 5:

“When the domestic price is **at or above** the world price, supply becomes **S + quota**.”

So the correct curve is a **kinked** curve:

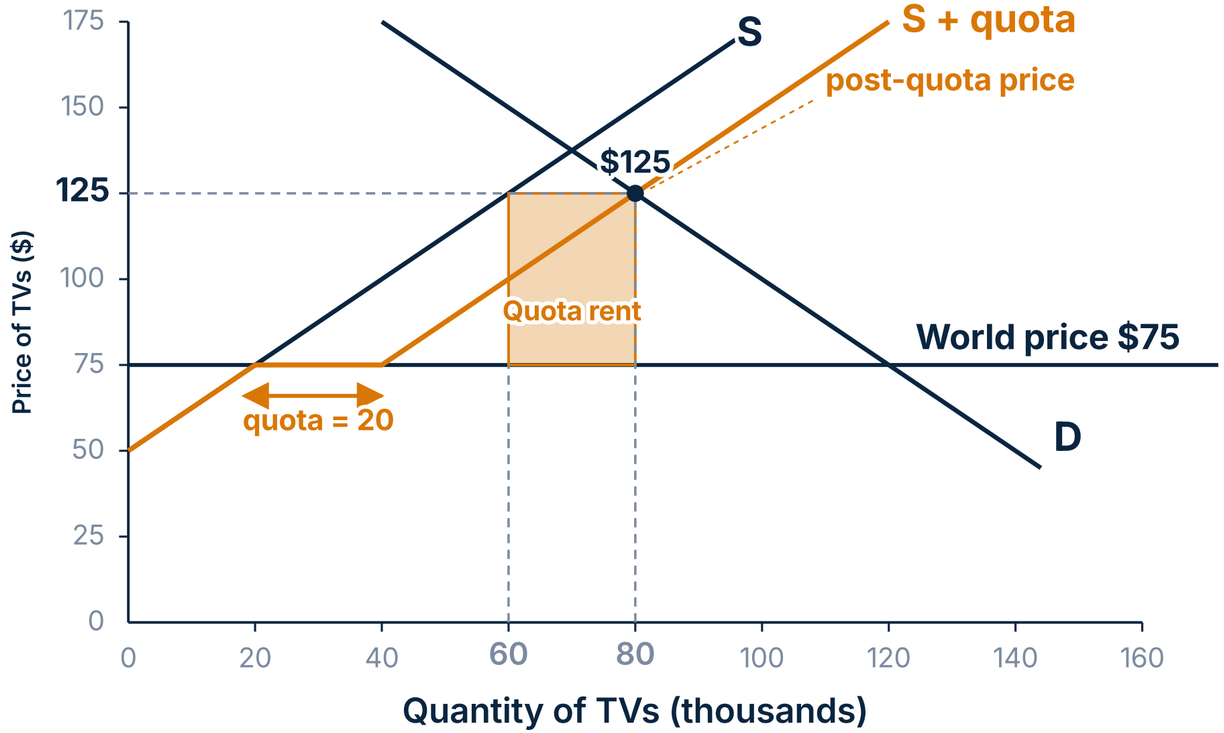
- **Below** $75 it **coincides with domestic S** — it does not run down to the price axis.
- **At** $75 it **jumps right** by the quota: 20,000 → 40,000 TVs.
- **Above** $75 it is domestic S **shifted right by 20,000**, starting at (40, $75).

The printed figure draws one straight line from about $25 on the price axis — that is the curve the question asks you to correct.

QUESTION 5.1 (VI)

Quota rent

FIG. 6 · THE QUOTA AT WORK



S + quota meets demand at $125: producers supply 60,000, imports are the 20,000 quota, and the rent rectangle is $1,000,000.

1. `S + quota` meets demand at `Q = 80` thousand, **P = $125**.
2. Domestic supply at $125 is 60 thousand, so imports = `80 - 60 = 20` thousand = the quota ✓
3. Quota rent = `($125 - $75) × 20,000`

> `Quota rent = $1,000,000`

The rent goes to the licence holders — the **importers**, not the government.

The handbag market in country A

5.2

QUESTION 5.2

Suppose country A is a small country where all buyers and sellers there are price takers. Below are the market demand and market supply equations of handbag in country A, which P is the price per handbag and Q is the number of handbags per year.

$Q_d = -2P + 40{,}000 \quad Q_s = 46P + 6{,}400$

a) Without international trade, what are the equilibrium price and equilibrium quantity of handbags in country A? Show all the steps of calculation.

Now country A is open for international trade. The world price per handbag is $600.

b) How many handbags does country A import under international free trade? Show all the steps of calculation.

To protect the local producers under international trade, country A government plans to impose tariff on imported handbags.

c) In order to decrease the import amount by 25% per year, what is the amount of per-unit tariff that the government should impose on imported handbags? Show all the steps of calculation.

d) What is the amount of tariff revenue? Show all the steps of calculation.

e) What is the amount of consumer surplus under tariff? Show all the steps of calculation.

f) “The imposition of tariff is beneficial to country A as a whole because the government can receive tariff revenue.” Explain whether the statement is true or false.

Setting up the algebra

Market demand and supply

$$Q_d = -2P + 40{,}000$$

$$Q_s = 46P + 6{,}400$$

P = price per handbag ($) · Q = handbags per year. Country A is a small country, so it takes the world price as given.

The method — the same three moves in every part

- **Autarky:** set $Q_d = Q_s$ and solve for P, then for Q.
- **Free trade:** evaluate both equations at the **world price** $P_W = \$600$.
- **Imports:** $Q_d - Q_s$ at the price buyers actually face. With a tariff, that price is $P_W + t$.

$$\text{Imports} = Q_d - Q_s = (-2P + 40{,}000) - (46P + 6{,}400) = \mathbf{33{,}600 - 48P}$$

QUESTION 5.2 (A) · WITHOUT INTERNATIONAL TRADE

Autarky equilibrium

1. “Without international trade” means the market clears domestically: $Q_d = Q_s$.
2. $-2P + 40{,}000 = 46P + 6{,}400$
3. $40{,}000 - 6{,}400 = 46P + 2P \rightarrow 33{,}600 = 48P$
4. $P^* = 33{,}600 / 48 = \mathbf{\$700}$
5. Substitute back: $Q^* = -2(700) + 40{,}000 = \mathbf{38{,}600}$
6. Check with supply: $46(700) + 6{,}400 = 38{,}600$ ✓

EQUILIBRIUM PRICE

$700

EQUILIBRIUM QUANTITY

38,600

The two curves cross at $700 — the price country A would have with no trade at all.

QUESTION 5.2 (B)

Imports under free trade (P_w = $600)

1. Quantity demanded at $600: Q_d = -2(600) + 40,000 = **38,800**
2. Quantity supplied at $600: Q_s = 46(600) + 6,400 = **34,000**
3. Imports = 38,800 - 34,000
4. Short cut with the import function: 33,600 - 48(600) = 4,800 ✓

> Free-trade imports = **4,800 handbags per year**

The world price $600 is below the autarky price $700, so A imports.

Q_D AT $600
38,800

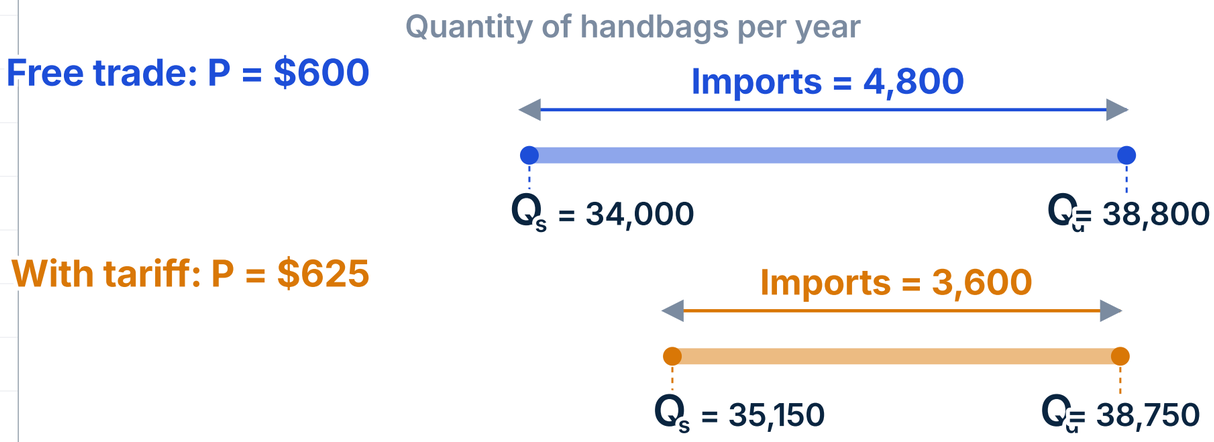
Q_S AT $600
34,000

IMPORTS
4,800

QUESTION 5.2 (C)

The tariff that cuts imports by 25%

FIG. 8 · Q5.2: IMPORTS BEFORE AND AFTER THE TARIFF



Reduction in imports = 1,200 → DWL = ½ × $25 × 1,200 = $15,000

A $25 tariff moves the domestic price from $600 to $625 and cuts imports from 4,800 to 3,600 handbags a year.

1. Target imports = `4,800 × 0.75 = 3,600`.
2. Find the domestic price that gives 3,600: `33,600 - 48P = 3,600`
3. `48P = 30,000` → `P = $625`
4. The tariff is the gap between the domestic price consumers pay and the world price: `t = 625 - 600`

> **`t = $25 per handbag`**

At $625: Q_d = `38,750`, Q_s = `35,150`, imports = `3,600` ✓

QUESTION 5.2 (D)

Tariff revenue

Tariff revenue is the tax per imported unit multiplied by the number of units still imported — a rectangle, not a triangle.

1 Revenue = $t \times (Q_d - Q_s)$

2 $= \$25 \times 3{,}600$

Tariff revenue = $90,000 per year

TARIFF PER UNIT	IMPORTS AFTER TARIFF
$25	3,600

Where the money goes

The government collects the price gap $625 - $600 on each of the 3,600 handbags that are still imported. This is the one gain that makes a tariff look attractive — part (f) shows why it is not enough.

QUESTION 5.2 (E)

Consumer surplus under the tariff

1. Quantity demanded at the post-tariff price \$625: $Q_d = -2(625) + 40{,}000 = \mathbf{38{,}750}$
2. The demand curve hits the price axis where $Q_d = 0$: $P = 40{,}000 / 2 = \mathbf{\$20{,}000}$ (the choke price).
3. Base = 38,750 handbags; height = $20{,}000 - 625 = \$19{,}375$
4. $CS = \tfrac{1}{2} \times 38{,}750 \times 19{,}375$

$CS = \mathbf{\$375{,}390{,}625}$

For comparison

PRICE CONSUMERS PAY	CONSUMER SURPLUS
\$600 (free trade)	\$376,360,000
\$625 (with tariff)	\$375,390,625
Loss to consumers	**-\$969,375**

A \$25 tariff costs consumers about **\$969,375** — but the government collects only \$90,000 and producers gain \$864,375. The rest, \$15,000, is lost for good.

QUESTION 5.2 (F)

“A tariff is beneficial because the government receives revenue” — true or false?

False — for country A as a whole

GROUP	CHANGE IN WELFARE
Consumers	-$969,375
Domestic producers	+$864,375
Government (tariff revenue)	+$90,000
Country A, net	**-$15,000**

The gains are real, but they are **transfers**: consumers lose more than producers and the government together receive. The $15,000 nobody gets is the **deadweight loss**.

DWL = the two triangles

DWL = ½ × t × Δimports	= ½ × $25 × (4,800 - 3,600)
= ½ × $25 × 1,200	DWL = **$15,000**

- **Production distortion** (½ × 1,150 × 25 = $14,375): resources are pulled into handbag production that could have been bought abroad more cheaply.
- **Consumption distortion** (½ × 50 × 25 = $625): some mutually beneficial trades no longer happen.

So the statement is true only for **specific groups** — the government and domestic producers — and false for the **country as a whole**.

PART 4 OF 5 · QUESTION 5.3

The generator market in Egypt

QUESTION 5.3

The electric generator market in Egypt is represented by the following equations:

$Q_d = 10{,}000 - 10P \quad Q_s = 1{,}000 + 10P$

a) Suppose the current world price of electric generators is US$300. If the Egyptian government introduces a US$100 tariff on imported electric generators, what will be the price of an electric generator in Egypt? What will be the quantity of electric generators produced in Egypt and quantity imported into Egypt? What will the government's tariff revenue be?

b) Who will gain and who will lose due to the above tariff? Calculate the changes in their surpluses. What will be the deadweight loss due to the tariff? Illustrate the situation with a diagram.

Price, output, imports and tariff revenue

Given

$$Q_d = 10{,}000 - 10P$$

$$Q_s = 1{,}000 + 10P$$

World price $P_W = \$300$; tariff $t = \$100$ per generator.

Steps

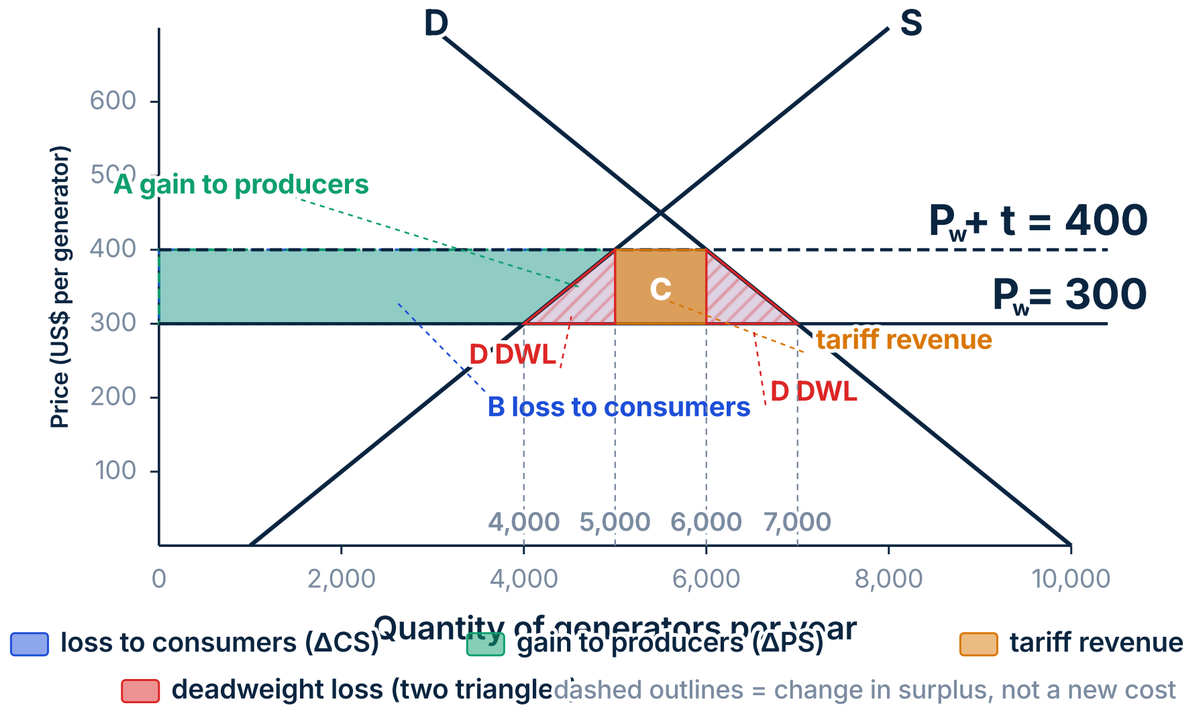
1. Price in Egypt = $P_W + t = 300 + 100 \rightarrow$ **$400**
2. Domestic production: $Q_s = 1{,}000 + 10(400) =$ **5,000**
3. Domestic demand: $Q_d = 10{,}000 - 10(400) =$ **6,000**
4. Imports = $6{,}000 - 5{,}000 =$ **1,000**
5. Revenue = $\$100 \times 1{,}000 =$ **$100,000**

$400	5,000	1,000	$100,000
PRICE WITH TARIFF	PRODUCED IN EGYPT	IMPORTED	TARIFF REVENUE

QUESTION 5.3 (B)

Who gains, who loses, and the deadweight loss

FIG. 9 · Q5.3(B): WHO GAINS AND WHO LOSES



The tariff transfers income to producers (A) and the government (C); the two hatched triangles are the $100,000 deadweight loss.

GROUP	CHANGE	AREA
Domestic producers	+$450,000	A
Egyptian government	+$100,000	C
Consumers	-$650,000	B
Country as a whole	-$100,000	D

1. **ΔPS**: ½(4,000 + 5,000) × $100 = **$450,000**
2. **ΔCS**: -½(7,000 + 6,000) × $100 = **-$650,000**
3. **DWL**: ½ × 1,000 × $100 × 2 = **$100,000**

PART 5 OF 5 · QUESTION 5.4

A recent development in international trade

QUESTION 5.4

Select one newspaper or magazine article (no earlier than year 2021) about the recent development in international trade or globalization. Briefly introduce the article content. Then explain how the knowledge that you have learnt in Topic 5 can be applied in the selected article.

Our article: Blenkinsop, P. (2024) "EU slaps tariffs on Chinese EVs, risking Beijing backlash", Reuters, 29 October 2024.

The EU's 2024 duties on Chinese electric vehicles

The article

Blenkinsop, P. (2024) *"EU slaps tariffs on Chinese EVs, risking Beijing backlash"*, Reuters, 29 October 2024.

After a year-long anti-subsidy investigation, the EU imposed **definitive countervailing duties for five years**, on top of the standard 10% duty:

BYD	17.0%	Tesla (Shanghai)	7.8%
Geely	18.8%	Other cooperating firms	20.7%
SAIC	35.3%	Total duty, some models	up to 45.3%

EU members were divided; Beijing opened its own probes into EU brandy, dairy and pork.

Mapped to Topic 5

TOPIC 5 CONCEPT	IN THE ARTICLE
Barrier type	An **ad valorem tariff** — a % tax on imports
EU consumers	Pay more for cars → consumer surplus falls
EU producers	Protected from cheaper imports → producer surplus rises
EU budget	Collects the duty on every imported car (tariff revenue)
Efficiency	Fewer cars imported → the two DWL triangles
Quota rent again	If a **minimum import price** replaces the duty, the price gap goes to the **exporters** — the VER case in Unit 5
Comparative advantage	Production stays where costs are higher

Three things to take away

1 · Both barriers raise the price

A tariff fixes the price gap (t); a quota fixes the quantity and lets the price rise to the level that clears the market with those imports.

2 · Someone always gets the gap

Tariff → the government. Quota → the licence holder (quota rent). If the rent goes abroad, the country is worse off still.

3 · Producers win, consumers lose, the country loses

The gains are transfers. What is not transferred — the two **DWL triangles** — is the net cost: $15,000 in Q5.2, $100,000 in Q5.3.

Protection can help specific groups, but it does not make the country as a whole better off — the question is always *whose* surplus changes, and by how much.

References and Q&A

Article (Q5.4)

- Blenkinsop, P. (2024) “EU slaps tariffs on Chinese EVs, risking Beijing backlash”, *Reuters*, 29 October 2024. reuters.com

Supporting: European Commission (2024) *Commission Implementing Regulation (EU) 2024/2754* and press release *IP/24/5589*, “EU imposes duties on unfairly subsidised electric vehicles from China”, 29 October 2024.

Data and figures

- Country Z and country A data: tutorial question sheet, Q5.1 and Q5.2.
- Egypt data: tutorial question sheet, Q5.3.
- All diagrams redrawn from the assignment figure and Unit 5 (Figures 5.4 and 5.6) — same notation, same colour code.

Course sources

- *AF3625 Topic 5 Lecture Slides* — free trade, tariffs, quotas, quota rent, winners and losers.
- *AF3625 Topic 2 Lecture Slides* and *Unit 2* — consumer surplus, producer surplus, social surplus.
- *AF3625 Topic 1 Lecture Slides* — demand, supply and equilibrium.
- *Unit 5 (with answers)* — worked tariff and quota figures, Figure 5.6 (S').

Thank you — questions?

Likely questions: why the quota rent is not government revenue · is a tariff ever good for the country · what changes if the country is *large*?